Matrix Factorization & Latent Embeddings via SVD for Recommendation

CS 550: Massive Data Mining — Semester Project

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MACHINE LEARNING | RECOMMENDER SYSTEMS | SVD | MOVIELENS 1M

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Project Overview

What We Built & How We Evaluated It

Core Objectives

Built two recommender systems from scratch using only **NumPy/SciPy primitives** — no out-of-the-box monolithic recommender libraries.

Dataset: **MovieLens 1M** — 100,000+ ratings across 610 users and ~9,700 movies.

Evaluation: strict **80/20 per-user holdout split** to prevent data leakage — not a random global split.

Central Question

Can we prove that SVD's latent structure is simultaneously **powerful and exploitable**?

Beyond accuracy: addressed three Trustworthiness rubric options — **Option A** (Transparency), **Option C** (Controllability), and **Option E** (Robustness) — demonstrated via an interactive Streamlit dashboard.

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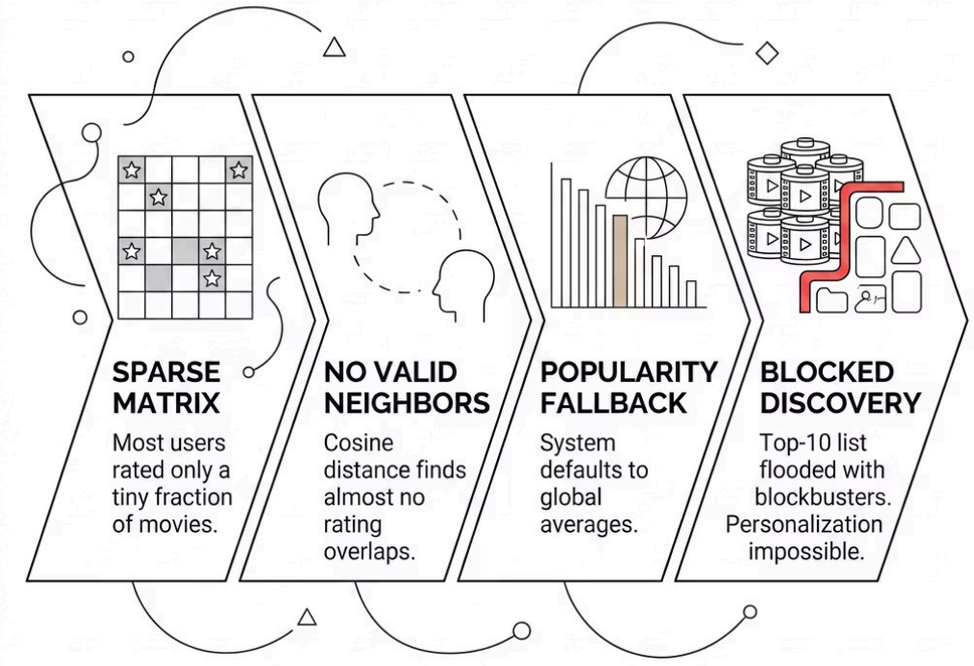
The Problem

Sparsity & Popularity Bias: Why Naive Recommendation Systems Break

The user-item rating matrix is **overwhelmingly empty** — most users have rated only a tiny fraction of ~9,700 available movies. This creates a cascade of failures that makes personalized discovery nearly impossible.

⚠ This is the **Cold Start + Popularity Bias trap**, and it demands a fundamentally different mathematical approach.



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Algorithm 1

Memory-Based Collaborative Filtering — The Baseline

How It Works

For each unrated movie, compute the **Pearson-Cosine similarity** between the target user and every user who has rated it. Prediction is a weighted average of neighbor ratings:

$$sim(u,v) = (r_u \cdot r_v) / (||r_u|| \, ||r_v||)$$

Conceptually simple and interpretable, but **structurally fragile** under data sparsity. When no valid neighbors exist, the system has no fallback — it collapses into useless popularity-based averages.

MAE Score

0.8173

Above the critical single-star variation threshold

NDCG@10

0.0047

Near-total failure on personalized ranking

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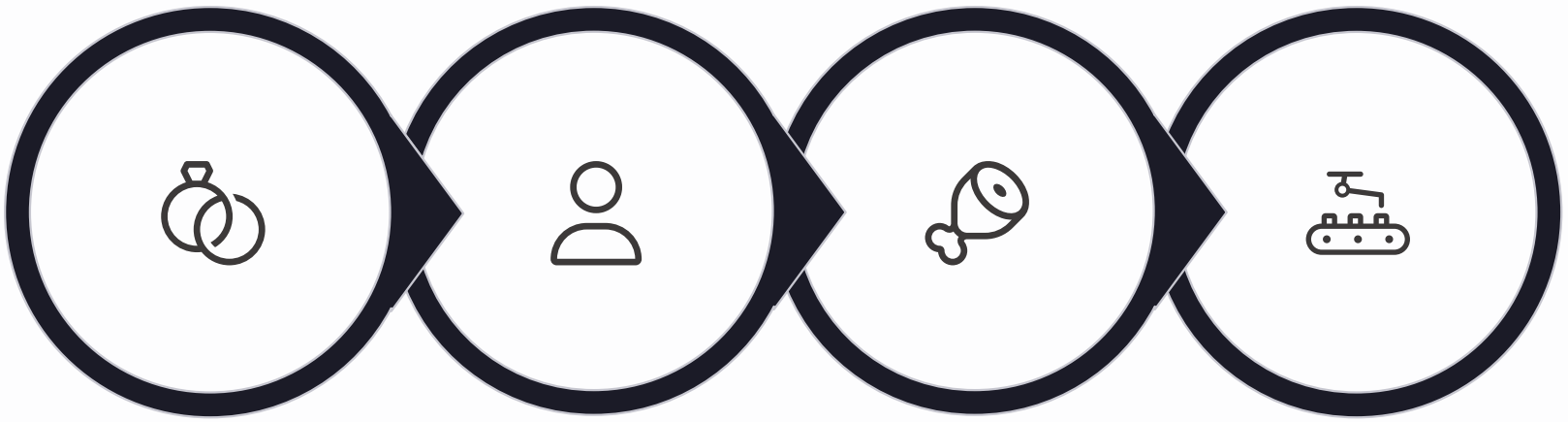
Algorithm 2

Model-Based SVD — Compressing the Matrix into Latent Dimensions



Sparse Input | Decompose k=50 | Add Global Mean | Dense Predictions

SVD decomposes the sparse matrix into **k = 50 dense orthogonal latent dimensions**, encoding hidden user taste patterns and hidden movie characteristics simultaneously. By operating in latent space rather than raw rating space, SVD naturally escapes the popularity trap and surfaces obscure, long-tail content — acting as an AI "smoothing" function that infers relationships even for movies with near-zero interaction history.

0.7262

MAE

Below the critical single-star threshold — dramatic improvement

0.4643

NDCG@10

Personalized ranking quality — a transformational leap

50

Latent Dims

Dense orthogonal dimensions encoding taste & movie characteristics

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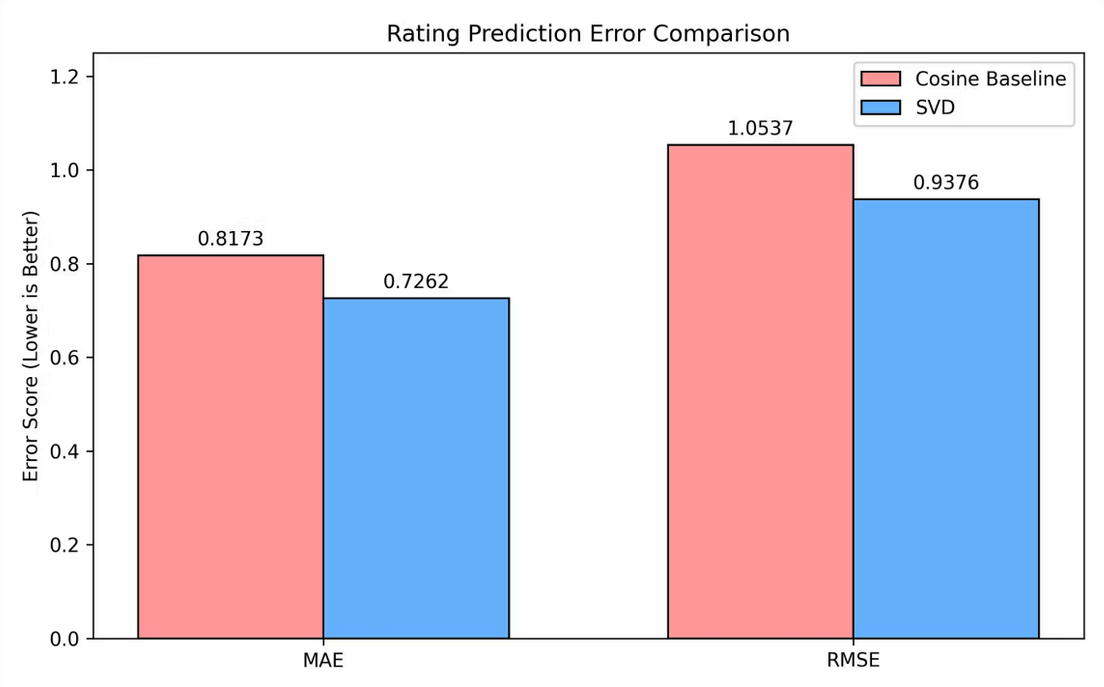
Results

Rating Prediction Accuracy: MAE & RMSE Comparison



Key Takeaways

- **MAE:** SVD 0.7262 vs. Baseline 0.8173 — SVD drops below the critical single-star variation threshold; the Baseline does not
- **RMSE:** SVD 0.9376 vs. Baseline 1.0537 — SVD stays under 1.0, meaning worst-case errors are meaningfully smaller
- The Baseline's high MAE directly reflects the **Cold Start collapse** — when neighbors don't exist, every prediction falls back to a useless global average
- SVD's 50-dimensional latent compression acts as a **natural noise filter**

Lower is better on both axes — **SVD wins on every rating accuracy metric**

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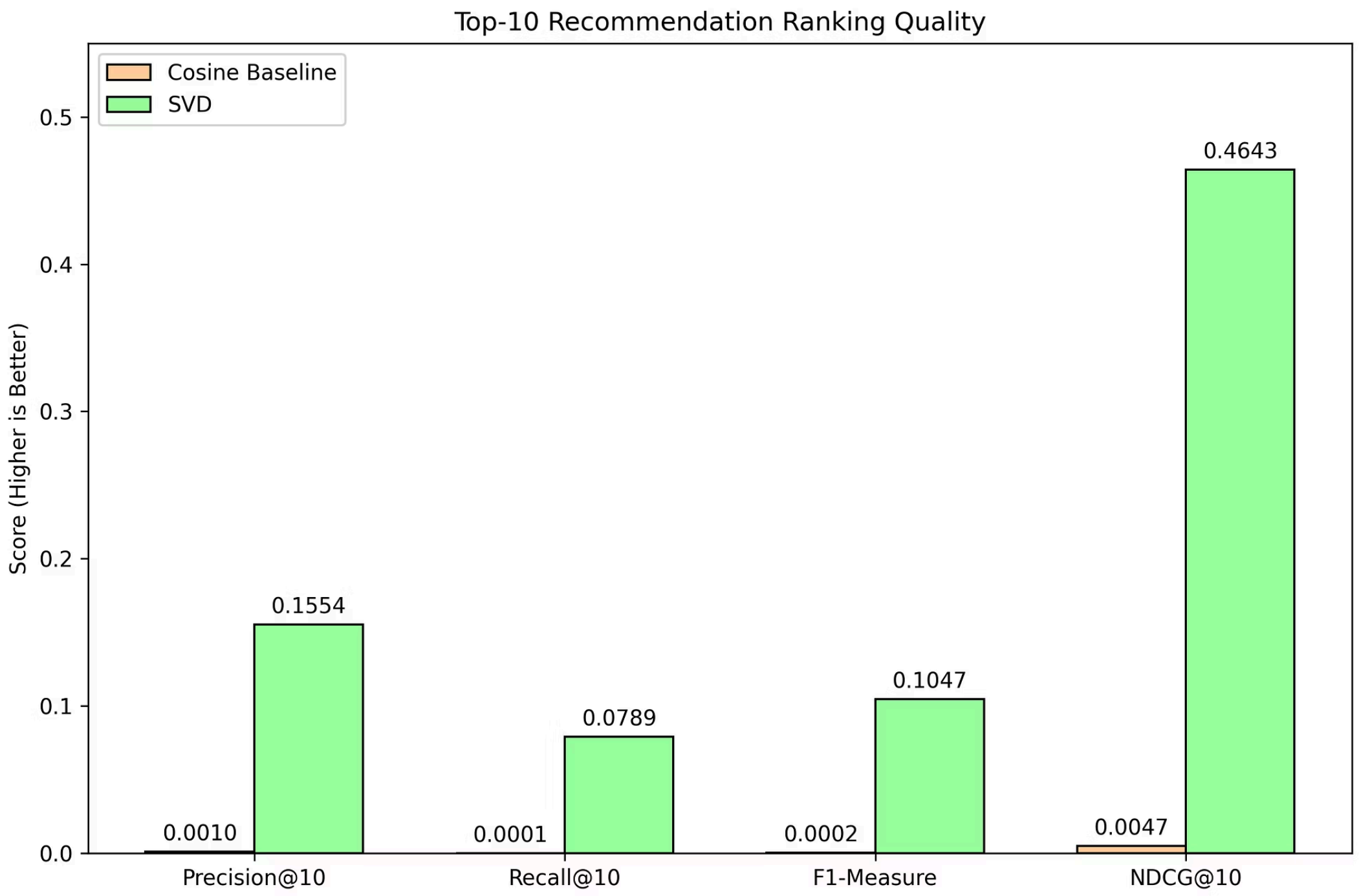
Results

Top-10 Ranking Quality: Precision, Recall, F1 & NDCG



The Baseline bars are nearly invisible across all four metrics — this is a **structural failure of Memory-Based CF**, not a marginal gap. SVD's Top-10 lists contain ~155x more relevant movies per recommendation served (Precision@10), and NDCG@10 shows a **9,757% improvement** — NDCG heavily penalizes any system that fails to rank the most relevant items at the very top. The F1-Measure confirms SVD's gains are balanced, not inflated by one metric alone.

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Trustworthiness — Option A

Transparency & Explainability: Cracking Open the Black Box

The Problem

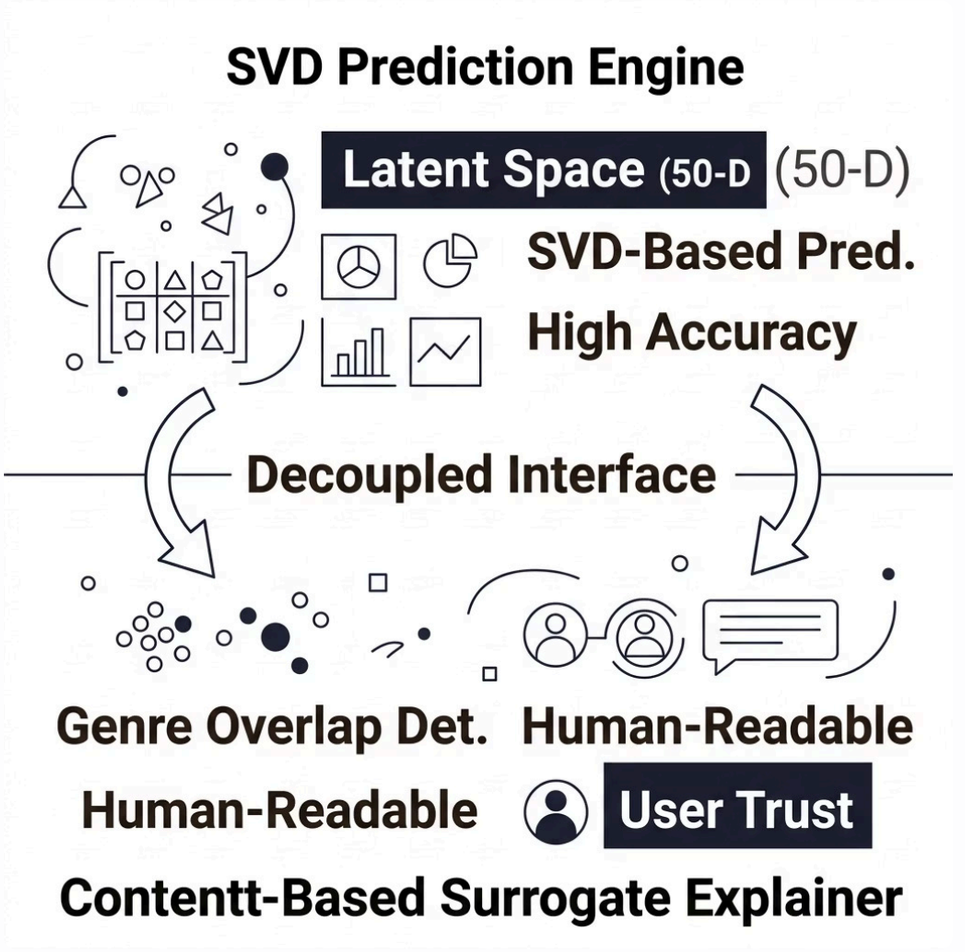
Raw 50-dimensional latent vectors are **completely uninterpretable** — SVD is a mathematical black box that users fundamentally distrust. There is no intuitive way to explain why a particular movie was recommended from raw latent coordinates alone.

The Solution

A **hybrid Surrogate Explanation architecture** — SVD handles prediction; a separate Content-Based layer generates the human-readable justification. The surrogate automatically identifies genre overlaps between the SVD recommendation and the user's historical favorites.

> UI output: *"Because you enjoyed [Movie X], our AI found strong thematic overlaps here."*

Prediction and explanation are **fully decoupled** — the SVD engine is never compromised for the sake of interpretability.



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Trustworthiness — Option C

Controllability: Putting the User Back in Control

🚫 The Problem

Traditional Matrix Factorization gives users **zero ability to steer or override** what the algorithm serves them. The system is a closed loop — opaque, unresponsive, and paternalistic by design.

🎛️ The Solution

An **interactive genre-filter dropdown** built directly into the Streamlit UI. Users can globally strip specific genres (e.g., "Remove all Sci-Fi recommendations") from the active SVD output in real time.

⚙️ Technical Integrity

The filter applies **post-prediction**, fully preserving the mathematical integrity of the underlying SVD tensor. The latent space is never altered — only the final output layer is filtered.

🏭 Industry Alignment

Mirrors **intent-driven steering modules** used in commercial streaming applications — trustworthiness requires giving users meaningful, transparent agency over algorithmic outputs.

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Trustworthiness — Option E

The Targeted Clone Swarm Attack: Designing an Advanced Shilling Attack

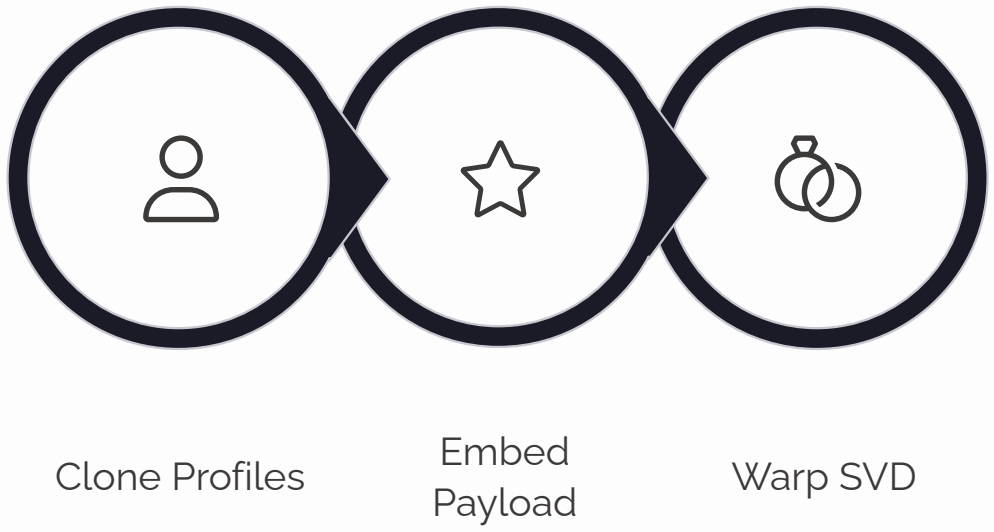
Why Standard Attacks Fail

Standard bot attacks inject **random noisy ratings** — SVD naturally filters these out as statistical noise because they don't align with any coherent latent behavioral cluster.

Our Novel Attack Vector

A **Targeted Clone Swarm of 500 bots** engineered to perfectly replicate one victim user's exact viewing history. Each cloned profile embedded a single maximum **5-star rating** for a low-profile payload movie (e.g., *Catwoman*).

500 clones + 1 victim = **501 statistically identical profiles** suddenly dominating the entire user cluster. When SVD retrains on the poisoned matrix, it is mathematically compelled to warp its k=50 principal dimensions — the target movie rises sharply into the victim's Top-10.



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SLIDE 11

The Mathematical Collateral Damage

The most significant discovery from poisoning SVD: the attack did not only elevate the intended target movie — it **violently reshuffled the entire global latent ecosystem.**

🎯 The Core Finding

Because SVD globally reconfigured its **50 dimensions** to minimize variance against the clone swarm, completely unrelated, un-targeted movies were dragged into the victim's Top-10 as well.

🏴‍☠️ Collateral Movies

Movies like *Pirates of the Caribbean* were **never targeted** — they simply shared vague latent structural DNA with the victim's profile and got caught in the gravitational pull of the tensor warp.

⚠️ Production Safety Implication

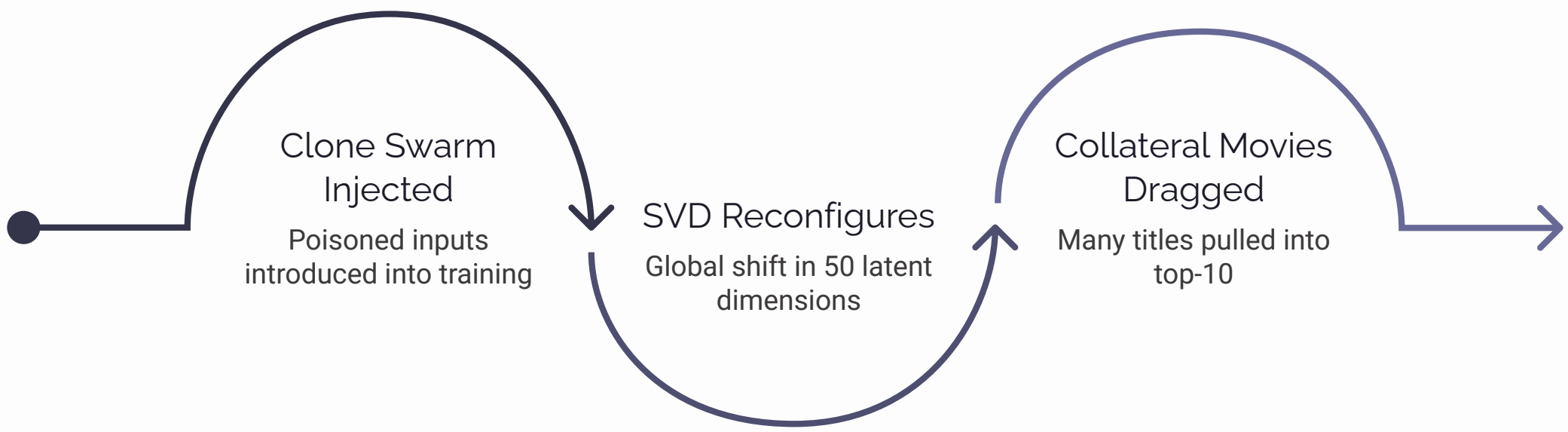
Any recommender system built on matrix factorization is **structurally vulnerable** to this class of coordinated attack — poisoning is not surgical, it is systemic.

Core Proof

Poisoning an SVD matrix doesn't surgically elevate one movie.

> It violently reshuffles the entire global latent ecosystem — every dimension, every embedding, every recommendation.

⊗ Mathematical Collateral Damage is a **verifiable, novel security risk** for any production recommender system built on matrix factorization.



The diagram above illustrates how a targeted clone swarm injection cascades through SVD's global compression mechanism, producing unintended collateral damage far beyond the original attack target.

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SLIDE 12

Conclusions & Key Takeaways

What We Built. What We Proved. A comprehensive evaluation of SVD-based matrix factorization — its superiority, its trustworthiness gaps, and its structural vulnerabilities.

SVD Dominance

SVD outperforms Memory-Based CF on **every single metric** — latent dimensionality is the correct and proven solution to Cold Start, sparsity, and popularity bias problems.

Trustworthiness Rubric

Options **A, C, and E** address complementary gaps — interpretability, user agency, and adversarial robustness. **No single option is sufficient alone.**

The Dual-Edged Sword

SVD's greatest mathematical strength — **global latent compression** — is simultaneously its greatest structural vulnerability when facing coordinated clone swarm attacks.

Verified Security Risk

Mathematical Collateral Damage is not theoretical — it is a **verifiable, novel security risk** for any production recommender system built on matrix factorization. The Targeted Clone Swarm attack demonstrates this conclusively.

⚠ No matrix factorization system should be deployed in production without adversarial robustness evaluation.

Future Work

01

Differential Privacy

Apply differential privacy constraints on SVD training to limit the influence of injected clone profiles.

02

Adversarial Retraining

Retrain models on poisoned matrices to build structural resistance to clone swarm injection attacks.

03

Explainability Layer

Extend the surrogate explainability layer to produce **user-facing confidence scores** for every recommendation.

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